Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Full-stack software engineer with 2+ years building responsive applications, REST APIs, microservices, and cloud workflows. Experienced in React, Angular, TypeScript, Python, Azure, SQL, Docker, Kubernetes, CI/CD, automated testing, observability, and Agile delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected Azure SQL and Blob Storage services for 500+ users, refining normalized schemas, indexed queries, and secure access patterns to reduce member-facing dashboard latency from 1.2 seconds to 350 milliseconds.
- Refactored fault-tolerant background workflows for 1,000+ jobs using explicit states, retries, and failure isolation, tripling concurrent throughput while improving maintainability, operational observability, and reliable production support.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed responsive React components and Flask REST services for secure upload, routing, status, and history workflows, translating user requirements into maintainable features supporting 150+ users and 10,000+ daily requests.
- Created concurrent processing for 1,000+ files with automated validation, error handling, tests, and AWS RDS persistence, reducing turnaround 60% while maintaining stable application behavior during peak workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible data and simulation pipelines processing 100,000+ records, applying structured automated testing, debugging, and performance analysis to improve long-horizon planning accuracy 18% across controlled evaluations.
- Participated in cross-functional collaborative design reviews and validation experiments, documenting technical decisions and tuning a spatial-temporal architecture that improved closed-loop stability 15% across repeated controlled simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed object-oriented API and ETL components using reusable software design patterns, AWS Glue, and S3, improving response times 30% and pipeline efficiency 68% across complex customer workflows.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating defect diagnosis and improving engineering confidence in production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular MERN services and WebSocket synchronization across eight customer, billing, scheduling, permissions, and notification workflows, supporting 2 million+ Azure SQL records with independently maintainable service boundaries.
- Automated Docker builds, tests, staged deployments, and production monitoring through Jenkins and Azure services, reducing release time 50% while improving overall software quality and consistent operational reliability.

Skills

Frontend & Languages: React, Angular, TypeScript, JavaScript, Python, Java, C++, HTML5, CSS3, Tailwind CSS, responsive UI, component design
Backend & Architecture: REST APIs, microservices, Flask, FastAPI, Node.js, WebSockets, object-oriented programming, JWT, API integration, asynchronous workflows
Data & Azure: Azure, Azure SQL, SQL Server, Blob Storage, PostgreSQL, MySQL, MongoDB, NoSQL, data modeling, indexing, query optimization, database design
Quality & DevOps: Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, PyTest, Mockito, unit testing, integration testing, code coverage, code review, debugging, observability, Agile, Scrum

Projects

HMIS | Live | *TypeScript, React, REST APIs, Tailwind CSS, Sentry*
- Built responsive appointment and member-style service workflows with consent-based API access, secure authentication, automated notifications, and Sentry observability, improving reliability and issue visibility across customer-facing digital experiences.

FinNexus | Live | *TypeScript, React, Tailwind CSS, WebSockets, Kubernetes*
- Built a responsive platform supporting 50,000+ users and one million transactions, integrating secure APIs, real-time synchronization, advanced filtering, MongoDB persistence, Kubernetes, and automated CI/CD deployment workflows.

PiXMod | Live | *Next.js, Angular, TypeScript, MongoDB, Stripe*
- Delivered a responsive full-stack SaaS application with Angular and Next.js interfaces, secure subscription workflows, REST integrations, and MongoDB persistence, reliably processing 1,000+ customer-edited digital media assets.